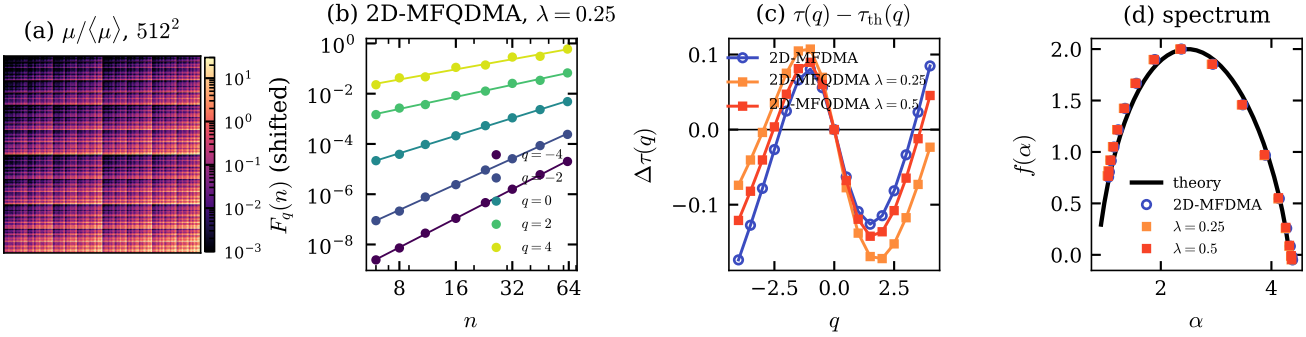

(a) $\mu/\langle\mu\rangle$, 512^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
(b) 2D-MFQDMA, $\lambda = 0.25$
$F_q(n)$ (shifted)
10^0
10^{-2}
10^{-4}
10^{-6}
10^{-8}
$q = -4$
$q = -2$
$q = 0$
$q = 2$
$q = 4$
8
16
32
64
n
(c) $\tau(q) - \tau_{\rm th}(q)$
$\Delta\tau(q)$
0.1
0.0
−0.1
2D-MFDMA
2D-MFQDMA $\lambda = 0.25$
2D-MFQDMA $\lambda = 0.5$
−2.5
0.0
2.5
q
(d) spectrum
$f(\alpha)$
2.0
1.5
1.0
0.5
0.0
theory
2D-MFDMA
$\lambda = 0.25$
$\lambda = 0.5$
2
4
α